Data Pricing for Blockchain-based Car Sharing: A Stackelberg Game Approach

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GLOBECOM 2020 - 2020 IEEE Global Communications Conference | 978-1-7281-8298-8/20/$31.00 ©2020 IEEE | DOI: 10.1109/GLOBECOM42002.2020.9322221

***Abstract*—With the increasing popularity of car sharing, a large amount of vehicle data has been generated which has great potential values for various applications (e.g., analyzing user habits for more economic benefits). These valuable data can be traded among owners and buyers on a data trading platform. Traditionally, data is traded in a centralized market which requires data exchange by trustworthy authorities. In this work, to address the potential unreliable issues (e.g., data loss and leakage), we design a consortium blockchain-based data trading framework to create a P2P trading market and enhance the security of data trading. We classify the data into five types to distinguish data with different values. Specifically, we investigate the pricing issue in the proposed car-sharing data market, which consists of data owner, service provider and data buyer. The data owner gives the pricing strategy of original data, and then the service provider processes the raw data and provides hierarchical quality of data with different data accuracy and privacy levels to the buyer who determines the data purchase strategy. Based on the interactions among these three parties, we formulate the problem as a three-layer Stackelberg game. Backward induction is applied to analyze the solution of the problem, and we conduct theoretical analysis to show the existence of Stackelberg game equilibrium. Numerical results evaluate the performance of our system under different settings.**

***Index Terms*—Car sharing, blockchain, data pricing, Stackelberg game.**

I. Introduction

Internet of Vehicles (IoV) have attracted enormous attentions from both academia and industry, and car sharing service has been a new trend in daily transportation. As car sharing becomes more and more popular in the transportation of smart cities, the massive amounts of data generated by car sharing service are collected by data owners. In practice, many car service providers, such as DiDi and MeiTuan, collect data (e.g., driving routes and user preferences) and take advantage of data analysis to improve their services for more economic benefits [1]. We consider that the data collected by car sharing belongs to the car sharing company, who has the right to process and sell the collected data. Obviously, to facilitate the exchange of data effectively and securely, the design of a reliable data trading platform for both data owners and buyers is required.

Traditionally, data is traded in a centralized market which requires data exchange by trustworthy authorities. However, centralized frameworks may bring potential unreliable issues, such as data loss, tampering data illegally or the leakage of user privacy. To address this issue, we apply blockchain technology [2] in the car sharing data market. Blockchain is distinguished by its characteristics of tamper proof, traceability and public accessibility, in which the consensus mechanism ensures synchronization. To this end, we design a consortium blockchain network to create a P2P trading market and enhance the security of data trading among data owner node, service provider node, buyer node and miners. Nevertheless, evaluating the real value of data in the blockchain-based data market is still challenging and will be discussed in this work.

Different kinds of data have different potential values due to different application purposes. Considering the characteristics of data in car sharing, we classify data into five categories according to the privacy level [3], which includes car driving information, In-car sensing data, car insurance data, leasing information and vehicle peccancy data. The data market consists of the data owner, service provider and data buyer, in which data owner collects raw data via car sharing service, and then service provider provides processed data with different level of accuracy to data buyer. To describe the data exchange among different parties and the corresponding strategic interactions, a three-layer Stackelberg game is formulated. Specifically, data owner is the leader who determines the pricing strategy of raw data, service provider in the middle layer sets the price for the processed data, and data buyer selects the data to buy and determines the size of purchased data.

The main contributions of this paper are as follows.

1) We apply consortium blockchain in the car sharing data market to create a trusted trading environment without the centralized intermediary and drive miners to maintain the blockchain security with financial incentives. Smart contract is built-in for executing the pricing and trading logic.
2) We formulate a three-layer Stackelberg game among data owner, service provider and data buyer to obtain an optimal pricing strategy. In this game, data owner is the leader to determine the pricing of raw data while service provider in the middle layer determine the pricing of processed data, and data buyer selects the data to buy and determines the size of purchased data.
3) We conduct numerical analysis to evaluate the performance of the proposed pricing mechanism and study the impact of different settings.

This work is supported by National Natural Science Foundation of China (Grant No. 61701230, 61801215), Natural Science Foundation of Jiangsu Province (Grant No. BK20170805).

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The rest of this paper is organized as follows. Section II presents the related work, and Section III introduces our system framework. In Section IV, we formulate the data trading problem as a three-layer Stackelberg game and study the optimal pricing strategy. Section V presents our numerical results and performance analysis. Finally, we conclude this work in Section VI.

II. Related Work

With the advancement of data analysis technology provided by machine learning and data mining, the value of data has become more and more important in many specific fields. Meanwhile, it has also attracted extensive research studies on data trading. However, unlike traditional commodities, the value of data is difficult to be estimated because of its virtual and abstract attributes. Various data pricing strategies have been proposed in order to figure out the real value and utility of data. In [4], data pricing models were categorized into two main groups: economic-based pricing model which was based on economic principles and game theory-based pricing model. Niyato *et al.* [5] took advantage of machine learning to obtain the utility function and applied Stackelberg game [6] to determine the optimal pricing.

To create a trusted environment in data trading, blockchain technology has been introduced in the data market recently. Liu *et al.* [7] proposed an edge/cloud-computing-assisted, blockchain-enhanced framework for data trading in IoT, where an authorized market-agency determined the winning owner and its optimal pricing by a two-stage Stackelberg game. In [8], a secure and decentralized IoV data-trading system based on blockchain was developed along with a debt-credit mechanism for enhancing the efficiency of data trading in IoV.

However, to sum up, it still lacks of a trusted P2P data trading framework and a corresponding data pricing mechanism particular for car sharing.

III. Blockchain-based Data Trading Framework for Car Sharing

In this section, we design a blockchain-based data trading framework that can address the challenges in car-sharing data market and build a data trading system model. Fig. 1 depicts the process of data exchange among different entities in the car sharing data market.

A. System Framework Overview

In the framework integrating blockchain network, data owner provides raw data for service provider who then processes data to provide services with different data accuracy. Data buyers buy data from service providers and can choose different level of data privacy and the size of data they want to purchase. These three parties form a game relationship in order to maximize their own interests.

Nodes in blockchain have their own accounts so that they can trade with each other without a third-party centralized platform. To build a transparent relationship for P2P transactions, we introduce consortium blockchain to construct a trusted environment, in which nodes are allowed to join in the blockchain after being audited. Note that there might exist few malicious nodes to attack the blockchain network such as DDOS or cyber attack. We adopt Proof-of-Work (PoW) consensus protocol to strengthen the security of blockchain.

We use smart contracts to execute the inherent logic of transactions, such as automatic pricing according to the data size requested by buyers. At the same time, the automation process of transaction has been improved, and the constraint of contract has also ensured the trust of parties in the transaction. When the transaction is completed, both parties evaluate each other, resulting in an impact on the credit score of both parties.

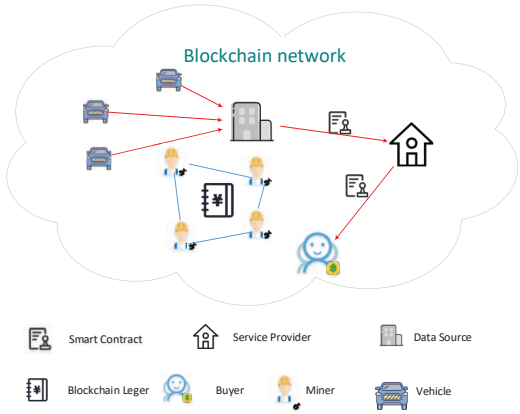

Fig. 1: System architecture of the blockchain-based car sharing data market.

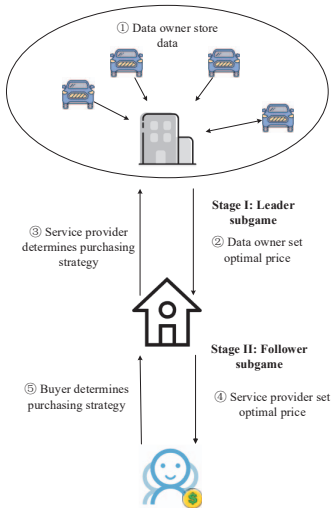

Fig. 2: Stackelberg game in the data market.

IV. Optimal Pricing for Car Sharing Data Trading

In the blockchain-based car sharing data market consisting of data owner, service provider and buyer, buyers pay data coins to sellers for obtaining the required data (raw data or processed data). This motivates us to further study the pricing issue among different entities. In this section, a three-layer Stackelberg game is formulated and the model is illustrated in Fig. 2.

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A. Problem Formulation

We consider a car sharing service application with various data generated and to be traded. The data is stored by the data owner, and the cost of transaction fee and packaging data is represented by c. p_s is the pricing of the original data, and only when $p_s > c$, the data owner will choose to sell the data. We denote the required data purchased from data owner by x, then the expected payoff of data owner can be obtained by

$$U_1(x, p_s) = (p_s - c) \cdot x, \tag{1}$$

where data is sold in the form of a data packet with different size x. We consider that data can be sold repeatedly such that x is theoretically infinite.

In the middle layer, after purchasing the original data, service provider needs to process the raw data to provide a better user service. The service provider provides N types of data accuracy that can be selected for users. The cost of processing data with different accuracy is different, and the higher accuracy is, the higher the cost will be. The set of data accuracy is denoted by $\mathbf{s} = \begin{bmatrix} s_1, & \ldots, & s_n \end{bmatrix}^\top$, so that we can denote the cost of processing unit data with accuracy i as

$$Q_i = qln(1 + s_i), \tag{2}$$

where q is a fitting parameter related to the cost of unit data. In addition to the cost of processing data, service provider also needs to pay transaction fees to enable transactions to be quickly verified. So the utility of service provider can be expressed as

$$U_2(x, p_b, p_s) = p_b x - Q_i x - f - p_s x, \tag{3}$$

where p_b denotes the pricing of processed data and f represents the transaction fee. Similarly, data will only be sold when the utility is positive, and we define the cost price for service provider be p_{bmin}, and p_{bmax} is the value when the utility of $U_2(x, p_b, p_s)$ equals to zero. So $p_{bmax} = \frac{A_j K_i ln(1+x)}{x}$, and when x tends to be infinitesimal, $p_{bmax} = A_j K_i$.

Data may have different values, and there are various ways to define different types of data. In this work, we consider that the data quality of car sharing is mainly related to the privacy level of the data, the accuracy of the data and the size of data. Therefore, we define the former five types of data in five different privacy levels which can be denoted by $\mathbf{r} = \begin{bmatrix} r_1, & \ldots, & r_5 \end{bmatrix}^\top$. We consider that the data quality of the privacy level j can be expressed as [3]

$$A_j = \alpha_1 - \alpha_2 e^{\alpha_3 r_j}, \tag{4}$$

where α_1, α_2, and α_3 are the curve fitting parameters of the utility function obtained by real-world experiments.

Besides the privacy level of data, quantity of data is also a main factor affecting the data quality. The data quality related to the quantity can be defined as [7]

$$D(x) = ln(1 + x). \tag{5}$$

The accuracy of the data also has a great impact on the quality of the data. Therefore, we define the impact of data accuracy on data quality as follows:

$$K_i = ks_i + m, \tag{6}$$

where parameters k and m are the fitting factors.

For the data buyers, they can determine the size of data with different accuracy and privacy level according to their demands, and thus the expected payoff of a buyer can be expressed as

$$U_3(x, p_b) = A_j \cdot D(x) \cdot K_i - p_b x. \tag{7}$$

B. Lower-level Stackelberg Game

We use backward induction to analyze the three-layer Stackelberg game. By analyzing the subgame between different layers, the Stackelberg equilibrium can be achieved. Intuitively, all parties adopt optimal pricing or purchasing strategy to maximize their own interests. Here we first analyze the game between buyer and service provider. For a given price p_b, the buyer decides the size of data packet to be purchased:

$$\max_{x, p_b} \quad U_3(x, p_b), \tag{8}$$

$$\text{s.t.} \quad x \geq 0. \tag{9}$$

By differentiating the equation $U_3(x, p_b)$, we have

$$\frac{\partial U_3(x, p_b)}{\partial x} = \frac{A_j \cdot K_i}{x + 1} - p_b. \tag{10}$$

After which, we can obtain the solution of x^* by solving $\frac{\partial U_3(x, p_b)}{\partial x} = 0$, and the result is

$$x^* = \frac{A_j \cdot K_i}{p_b} - 1. \tag{11}$$

Here, we consider the right-hand side of (11) to be positive. Otherwise, $x^* = 0$, and the problem becomes trivial.

The service provider can adjust data price p_b to maximize its utility according to the optimal purchasing strategy of the data buyer, and the subgame for service provider is as follows:

$$\max_{x, p_b} \quad U_2(x, p_b, p_s), \tag{12}$$

$$\text{s.t.} \quad x \geq 0 \tag{13}$$

$$p_b \in (p_{bmin}, p_{bmax}). \tag{14}$$

The purpose of this game is to set the optimal price given that the data buyer adopts the optimal strategy. Thus substituting $x^* = \frac{A_j \cdot K_i}{p_b} - 1$ into (3), equation (12) can be expanded as:

$$U_2(x^*, p_b, p_s) = (p_b - Q_i - p_s) \cdot (\frac{A_j \cdot K_i}{p_b} - 1) - f. \tag{15}$$

To find the Stackelberg equilibrium, we derive the first order derivatives of the $U_2(x, p_b, p_s)$ with respect to p_b, which is

$$\frac{\partial U_2(x^*, p_b, p_s)}{\partial p_b} = \frac{A_j \cdot K_i \cdot p_s}{p_b^2} + \frac{A_j \cdot K_i \cdot Q_i}{p_b^2} - 1. \tag{16}$$

Then the solution of p_b^* can be obtained by solving $\frac{\partial U_2(x^*, p_b, p_s)}{\partial p_b} = 0$, as follows:

$$p_b^* = \sqrt{A_j \cdot K_i \cdot (p_s + Q_i)}, \tag{17}$$

From (11), we know that $\frac{A_j \cdot K_i}{p_b^*} - 1 > 0$. So $p_s + Q_i < A_j K_i$, which means $p_b^* < A_j K_i$, and due to $p_{bmax} = A_j K_i$,

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we obtain $p_b^* < p_{bmax}$. Taking the second derivatives of $U_2(x, p_b, p_s)$, we have

$$\frac{\partial^2 U_2(x^*, p_b, p_s)}{\partial p_b} = -\frac{2 \cdot p_s \cdot K_i \cdot A_j}{p_b^3} - \frac{2 \cdot Q_i \cdot K_i \cdot A_j}{p_b^3} < 0, \tag{18}$$

which is obviously negative.

Therefore, we can conclude that since the second derivatives of $U_2(x, p_b, p_s)$ and $U_3(x, p_b)$ are both negative, x^* and p_b^* are both unique and globally optimal. In other words, x^* and p_b^* are the Stackelberg game equilibrium points.

C. Upper-level Stackelberg Game

In order to maximize the utility, the data owner can dynamically adjust the price of data according to the strategy of the service provider. Hence, the subgame for data owner can be formulated as:

$$\max_{x, p_s} \quad U_1(x, p_s), \tag{19}$$

$$\text{s.t.} \quad p_s > c. \tag{20}$$

Referring to the previous subsection, the Stackelberg equilibrium points can be expressed as:

$$p_s^* = arg \max_{x, p_s} U_1(x^*, p_s), \tag{21}$$

$$\text{s.t.} \quad (x^*, p_b^*) = arg \max_{x, p_b, p_s} U_2(x^*, p_b, p_s), \tag{22}$$

$$x^* = arg \max_{x, p_b} U_3(x, p_b). \tag{23}$$

Substituting x^* and p_b^* into equation (1) yields

$$U_1(x^*, p_s) = (p_s - c) \cdot \left(\frac{A_j \cdot K_i}{\sqrt{A_j \cdot K_i \cdot (p_s + Q_i)}} - 1\right). \tag{24}$$

Then taking the second derivatives of $U_1(x^*, p_s)$, we have

$$\frac{\partial^2 U_1(x^*, p_s)}{\partial p_s} = -\frac{A_j^2 \cdot K_i^2}{(A_j \cdot K_i \cdot (Q_i + p_s))(3/2)} - \frac{3 \cdot A_j^3 \cdot K_i^3 \cdot (c - p_s)}{(4(A_j \cdot K_i \cdot (Q_i + p_s)))(5/2)} < 0, \tag{25}$$

so p_s^* is unique and globally optimal.

In summary, we apply backward induction in the following $Algorithm1$ to obtain the optimal strategy x^* in the first place, and then the optimal strategy p_b^* of service provider is produced according to x^*. When the lower-level Stackelberg game reaches equilibrium points, we can obtain p_s^* by iteration. Overall, after running these algorithms, the game equilibrium is found.

V. NUMERICAL RESULTS

In this section, we conduct numerical analysis to evaluate the main factors which may affect the utilities of different parties in the blockchain. We analyze the impact of some main factors in the data market system by numerical calculations. We consider that the choices of data accuracy provided for buyer are $\mathbf{s} = [s_1, s_2, s_3, s_4, s_5]^\top$. The default parameters settings are as follows: the cost $c = 1$, fitting parameter $q = 1$, the transaction fee $f = 1$, data accuracy $\mathbf{s} = [0.1, 0.3, 0.5, 0.7, 0.9]^\top$, the privacy level factor $\mathbf{r} = [0.1, 0.3, 0.5, 0.7, 0.9]^\top$, fitting parameters of privacy level $\alpha_1 = 3, \alpha_2 = 0.1, \alpha_3 = 2$, and fitting parameters of data quality $m = 5, k = 5$.

We first choose $s = 0.5$ and $r = 0.5$. Then we obtain the optimal utility of data owner and the corresponding optimal size of purchased data as depicted in Fig. 3. We can clearly find that as the data pricing of data owner increases, the demand of data decreases. Given the curve of the utility of data owner,

Algorithm 1 Algorithm to find the Stackelberg game equilibrium points

Joining the blockchain network: Each node joins in the blockchain network and selects its role.
Initialize $p_s = p_s^{(0)}$, $p_b = p_b^{(0)}$, $x = x^{(0)}$, $xchange = 1$, $k = 0$ and the result accuracy $\varepsilon = 10^{-3}$.
while $xchange > \varepsilon$ **do**
  Buyer adopts strategy to obtain a higher utility:
  $x^{k+1} = \frac{A_j \cdot K_i}{p_b^k} - 1$,
  After the data buyer adopting its strategy, the service provider adjusts its own pricing strategy as:
  $p_b^{k+1} = \sqrt{A_j \cdot K_i \cdot (p_s^k + Q_i)}$,
  The data owner adjusts its pricing strategy:
  $p_s^{k+1} = arg \max_{p_s} U_1(x^{(k+1)}, p_s, p_b^{k+1})$,
  $k = k + 1$,
  $xchange = \left|x^{k+1} - x^k\right|$.
end while
The points finally obtained after the loop are the Stackelberg game equilibrium points in the data market.
Output: p_s^k, p_b^k, x^k.

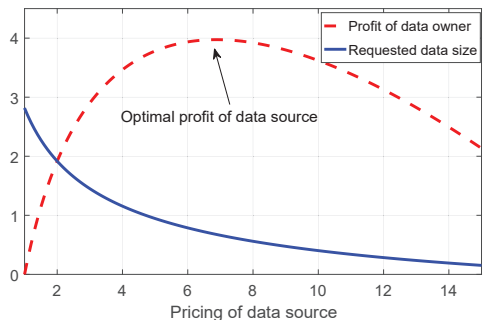

Fig. 3: Optimal requested data size and utility of data owner.

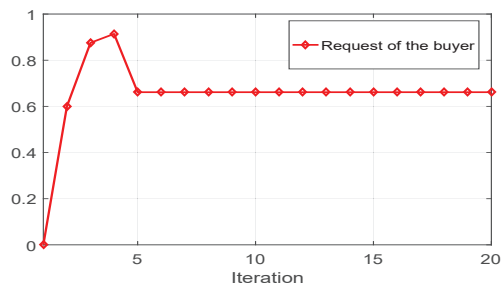

Fig. 4: Iteration of the requested data.

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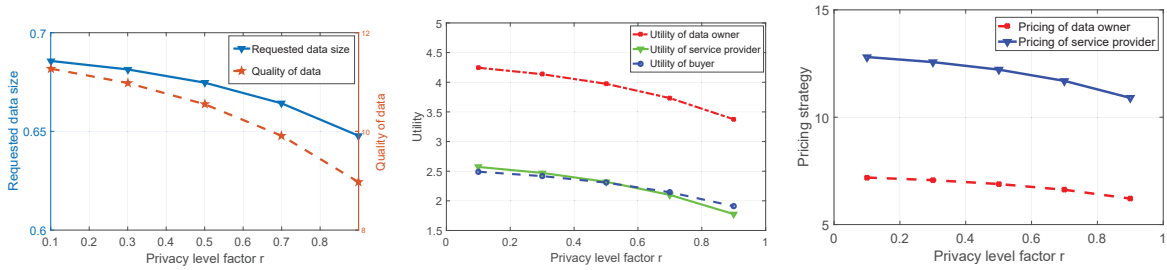
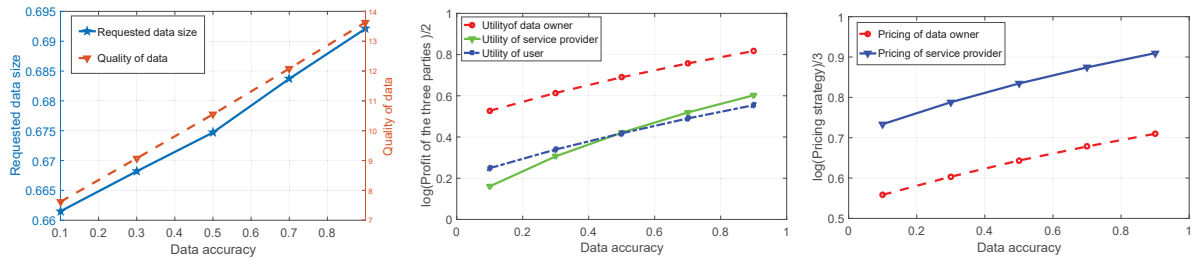


(a) Impact on the requested data size and the quality of data.

(b) Impact on utilities.

(c) Impact on pricing strategies.

Fig. 5: Impact of factor r.



(a) Impact on the requested data size and the quality of data.

(b) Impact on utilities.

(c) Impact on the pricing strategies.

Fig. 6: Impact of data accuracy.

we can obtain the optimal pricing when the service provider adopts the optimal purchasing strategy.

The solution of x^* is the equilibrium point of the Stackelberg game in the data market. Fig. 4 shows the convergence progress of the purchased data, and the initial parameters of the data owner, service provider, buyer are set as follows respectively: $x = 0.6$, $p_s = 4$, $p_b = 8$. We can find that the buyer's request converges quickly under the initial settings, and data buyers have a tendency to purchase more data packet to maximize the utility while after the leader adjusts its pricing strategy, the size of purchased data reduces quickly and finally converges to 0.66.

Fig. 5 shows the impact of privacy level and the falling tendency of the requested data size indicates that the Stackelberg equilibrium of the optimal purchasing strategy keeps decreasing. The three curves in subfigure (b) all have a downward trend, which means that the utilities of three parties decrease with the increase of r. In subfigure (c), the lower privacy quality of data means that the information of the data has less value, and to sell more raw data, it needs to reduce the price to attract more buyers.

The impact of data accuracy is shown in Fig. 6. When the buyer purchases the higher accuracy of data, the quality of data will rise significantly. Subfigure (b) shows that the utility of the data owner is the highest, and the utility of the buyer is higher than that of the service provider firstly while with the growth of the data accuracy, the utility of service provider becomes higher than that of the buyer. From subfigure (c), we can find that with the improvement of the data accuracy, both the pricing of data owner and service provider increase. For data owner, to maximize the utility, it tends to raise the price of the raw data when the requested data size is larger.

VI. Conclusion

In this paper, we study the blockchain-based data market for car sharing. Consortium blockchain and PoW mechanism create a trusted trading environment and smart contract is built-in to execute the pricing and trading logic. We formulate the data pricing problem as a three-layer Stackelberg game among data owner, service provider and data buyer. We apply backward induction to acquire the Stackelberg equilibrium. Both theoretical and numerical analysis have been conducted to evaluate the performance of the pricing mechanism.

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